

Abhinav Kataria

Full-Stack Developer | Python, Node.js & LLM Systems

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SUMMARY

Final-year Computer Science student who builds full-stack systems end to end. Built, as **sole engineer**, a **Python/FastAPI** backend at Jakson Group integrating an external **LLM** behind a PII-safe gateway and deterministic authorization pipeline, validated by **69 passing unit tests**. Built **31 of 108 commits** on NexRoute, a live **React + Node.js/Express + MongoDB** delivery-allocation platform, spanning frontend architecture, backend correctness fixes, and API security hardening. Strong **C++ DSA** base (500+ problems, **rank 1147**, LeetCode Biweekly 189). Rating: **1780+**

EDUCATION

Birla Institute of Technology, Mesra

2023 - 2027

B.Tech in Computer Science Engineering | **Coursework:** DSA, OOP, DBMS, Operating Systems, Computer Networks

Vishwa Bharati Public School, Noida

2008 - 2022

CBSE Class XII: **86.2%** | Class X: **95%**

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, C++, Java, C, SQL

Backend: FastAPI, Uvicorn, Node.js, Express, REST APIs & API Design, Socket.IO/WebSockets

AI & LLM Integration: LLM gateway design (PII-safe), sentence-transformers (all-MiniLM-L6-v2), embedding-based intent classification, cosine-similarity ranking, RAG grounding

Data: MongoDB/Mongoose, SQL & relational data access (pyodbc)

Security & Systems: JWT, OAuth 2.0, Role-Based Access Control, PII redaction & data minimization, API hardening (Helmet, CORS, rate limiting, mongo-sanitize), System Design

Frontend & Tooling: React, Next.js, Mapbox GL, Tailwind CSS, Git/GitHub, CI/CD, pytest, Vercel/Render

Also: PostgreSQL, Redis, Docker, AWS & Kubernetes/EKS

WORK EXPERIENCE

IT Intern, Jakson Limited

June 2026 - July 2026

Platform-Agnostic Enterprise AI Chatbot Engine - sole engineer

- Designed a **nine-stage synchronous request pipeline** (FastAPI + Uvicorn) placing deterministic authorization before any **LLM** call - the model phrases answers, never decides access.
- Architected a single neutral engine serving **multiple internal platforms** (LMS, HRMS) from one codebase using **8 JSON config files** per platform, with zero engine-code changes to onboard a new one.
- Built a **two-point, fail-closed** PII minimization layer - inbound redaction plus outbound allowlist field-projection - so no raw record reaches the external **LLM** provider.
- Implemented hybrid intent resolution over **local sentence-transformers embeddings** with confidence/margin gating, routing ambiguous queries into a bounded disambiguation loop instead of guessing.
- Cut avoidable LLM calls by routing FAQ/navigation queries through a deterministic knowledge path backed by **relational (SQL) lookups via pyodbc**.
- Validated the system with **69 automated pytest tests across 14 modules (100% passing)**.

PROJECTS

NexRoute - Real-Time Delivery Partner Allocation Engine

Live Demo | GitHub

- Built a full-stack real-time allocation engine (**React, Node/Express, Socket.IO, MongoDB**) matching orders to riders on a weighted model (ETA, rating, load), using **H3 hexagonal indexing** (res-7) to prune global scans into local lookups.
- Scaffolded the **React frontend** from scratch (Vite + React Router with protected routes) - app shell, layouts, navigation, and six core pages (Dashboard, Live Map, Orders, Riders, Settings, Allocation History).
- Built the **live map experience** with **Mapbox GL** rendering real-time rider positions streamed over **Socket.IO**, and fixed a bug that kept drawing routes for orders whose rider had already been reallocated.
- Fixed a manual-allocation correctness bug by filtering candidates to **IDLE-only riders** before scoring - mirroring the guard the automatic tick already applied.
- Hardened the **Express** backend with **Helmet, CORS allowlisting, rate limiting, and mongo-sanitize** middleware, plus fail-fast startup validation for required environment variables.
- Authored **31 of 108 commits**, including a **runtime orphan-order sweeper** that re-queues orders left stranded by a crashed or partial write instead of letting them stall or wrongly cancel.

POSITION OF RESPONSIBILITY & ACHIEVEMENTS

- Technical Lead, IEEE Student Chapter** and **Technical Member, GDSC**, BIT Mesra
- Technical:** Solved **500+ DSA problems in C++**; rank **1147** in LeetCode Biweekly Contest 189. Rating: **1780+**
- Academic:** Qualified JEE Main & Advanced 2023 (top 4%, category); cleared NTSE Level 1 (2019).